

Charlie Hoose

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SKILLS

- **Languages:** Python, R, SQL, C++
- **Libraries/Frameworks:** Pandas, NumPy, Matplotlib, seaborn, scikit-learn, LangChain, ggplot2, tidyverse, dplyr
- **Tools:** Tableau, Excel, PowerBI, Git/GitHub, LaTeX, Netezza, Kadince, R Markdown, Power Automate, Azure (Azure App Service, Azure ML, Azure CLI), Flask, Docker, SQLite
- **Methods:** Feature Engineering, ETL, Regression, Classification, Clustering, Statistical Inference, Experimental Design

WORK & RESEARCH EXPERIENCE

First Citizens Bank

Pasadena, CA

Data Analyst Intern

June 2025 – Present

- Leveraged SQL and Netezza to enhance philanthropy dashboards, joining new data fields to improve reporting accuracy
- Engineered multiple Excel VBA macros to automate dashboard color-coding for monthly reporting, reducing a 45-minute manual process to under 10 seconds with 100% accuracy for 5+ reports
- Streamlined Low-to-Moderate Income rate analysis by executing complex data joins and transformations in Excel using RegEx, saving \$250,000 in loan payments
- Developed an automated ETL reconciliation workflow in Excel to detect mismatching payments across tracking systems, saving 93% of manual processing time with 97% accuracy
- Led an intern team to analyze historic and projected philanthropy donations, creating and presenting Tableau dashboards to 60+ employees and informing a bank-wide philanthropic strategy

Italy National Baseball Team

Los Angeles, CA

Data Scientist

January 2026 – March 2026

- Trained an XGBoost model predicting pitcher-batter matchup outcomes on 2020–2024 MLB pitch-level data, achieving an R^2 of 0.40 and OPS bucketed correlation of 0.974 on 2025 holdout data
- Built an R Shiny app with player and team presets to generate exploitability grids identifying favorable matchups
- Delivered 7+ predictive dashboards and the R Shiny site to the Italian Team, directly informing World Baseball Classic strategy for batting lineups and pitching decisions

B. John Garrick Institute for the Risk Sciences at UCLA

Los Angeles, CA

Data Science Research Intern

April 2024 – July 2025

- Debugged and optimized AI-powered wildfire evacuation prediction application in VSCode to enable faster emergency planning and prediction for first responders and city planners
- Built an NLP pipeline to parse wildfire research papers, extracting keywords and contextual data to enhance model feature importance and cutting research synthesis time by over 80%
- Analyzed 5 competing wildfire evacuation platforms to identify performance gaps, presenting findings via matplotlib and seaborn visualizations to inform our team's modeling priorities
- Centralized findings across human and traffic evacuation model research into 3 reports, creating a referenceable knowledge base that informed I-Corps presentations

PROJECTS

Wildfire Acreage Prediction Project

April 2026

- Interviewed wildfire and GIS experts to source and validate national fire response datasets for more accurate modeling
- Performed data cleaning in pandas and ggplot2 visualizations to identify key trends influencing wildfire spread
- Constructed a wildfire size classification model using XGBoost, achieving 83% accuracy and 0.82 weighted F1 score
- Containerized wildfire prediction model with Docker and deployed as a live REST API endpoint on Azure App Service

Data Science Union Chatbot Project

March 2025

- Engineered a retrieval pipeline with LangChain, converting DSU website content into vector embeddings and ranking results via cosine similarity and max marginal relevance to user queries
- Designed a RAG chatbot to automate recruitment Q&A for DSU, enhancing information access for 750+ applicants
- Constructed an interactive Streamlit site for real-time user queries and contextualized answer generation using an LLM

Automated Resume Screener Project

April 2024

- Developed a Python script to parse candidate job histories and extract tenure-based features, improving model accuracy
- Conducted exploratory data analysis with matplotlib and seaborn to evaluate tenure distributions and feature patterns
- Built a KNN model with NLP-based features to automate resume screening, filtering 2,500 resumes down to 375

EDUCATION

University of California, Los Angeles (UCLA)

GPA: 3.9/4.0

B.S. in Statistics and Data Science, Minor in Data Science Engineering

- **Relevant Coursework:** Data Structures and Algorithms, Machine Learning, Deep Learning & Neural Networks, Statistical Modeling and Data Mining, Data Analysis and Regression, Relational Databases, Linear Algebra, Probability
- **Club Involvement:** Data Science Union (Data Scientist, Dir. of Membership, Executive Advisor), Statistics Club